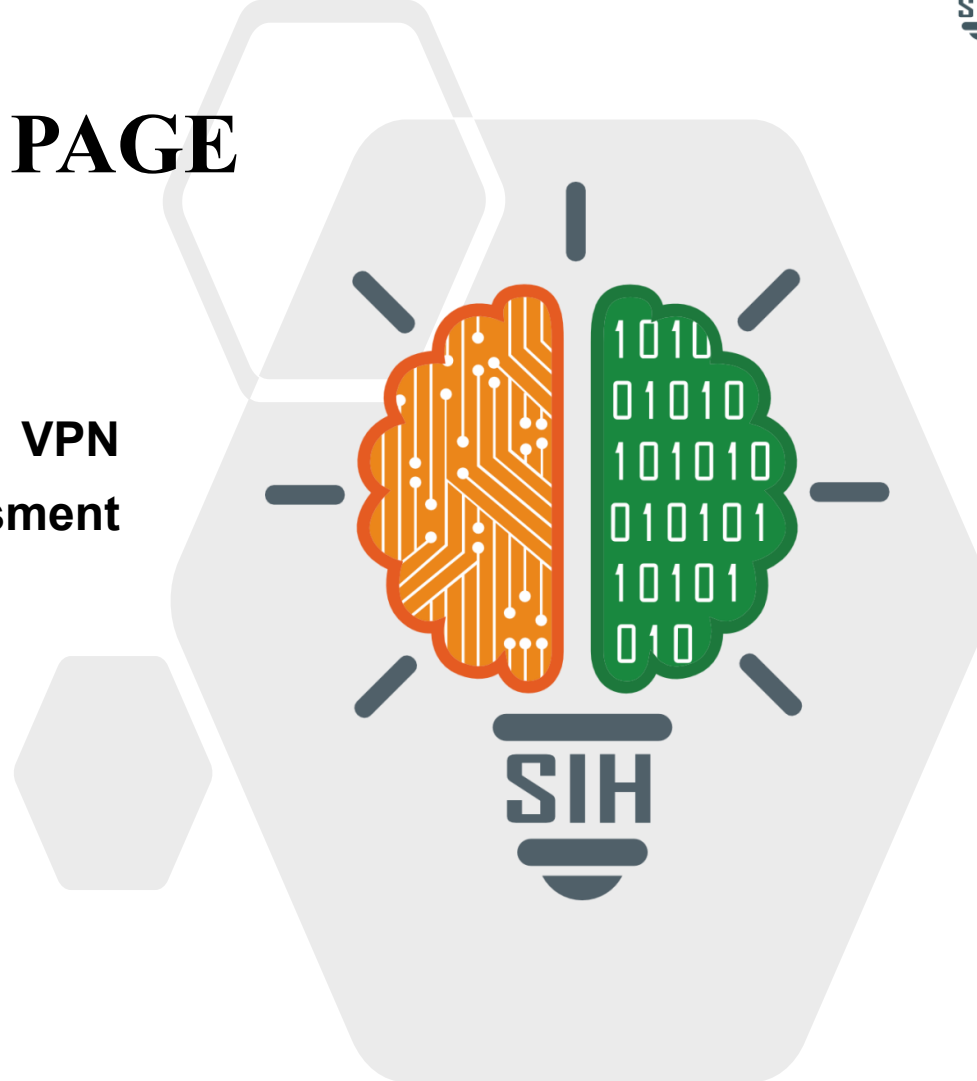


TITLE PAGE

- Problem Statement ID – SIH26160
- Problem Statement Title- AI-Powered IPsec VPN Protocol Analyzer & Security Assessment Framework
- Theme- Blockchain & Cybersecurity
- PS Category- Software
- Team ID- 183631
- Team Name- frontend-backend-tragedy



PROPOSED SOLUTION

- **One capture in, a scored assessment out.** VaultScope reads an IPsec PCAP or live stream and tells an analyst how strong the VPN is, with no manual packet inspection.
- **Two AI problems, kept separate.** Crypto identification is a cleartext parser, because IKE negotiates in the clear. Traffic type inside ESP is ML on packet-size and timing metadata.
- **Full assessment.** 21 CVE and RFC-linked rules, 5 compliance baselines, a metadata-exposure score, post-quantum readiness and attack detection that cites exact pcap frames.
- **Every output the brief asks for.** Risk score, threat matrix, AI confidence score, executive and technical reports, live dashboard, JSON/CEF export.
- **What is unique.** A hand-written RFC 7296 / 2408 decoder and a first-party labeled dataset from real strongSwan tunnels.

Two problems, deliberately not confused

Most tools treat VPN analysis as one task. It is two, and only one of them is machine learning.

THE HANDSHAKE IS IN CLEARTEXT

Crypto identification

Stage 2 — a parser, not a model

IKE negotiates in the clear before the tunnel exists. The cipher, DH group, PFS status and vendor are simply there to be read.

So we wrote a byte-level RFC 7296 decoder rather than training a model to guess at facts already on the wire.

ML enters only where the parse cannot: a truncated or malformed handshake falls back to Stage 4a, which recovers the DH group from message structure.

Result

Exact parameters, with a confidence source on every field — parser or classifier.

THE PAYLOAD NEVER DECRYPTS

Traffic-type inference

Stage 4b — the genuine ML problem

ESP payloads are encrypted and stay that way. We never attempt decryption and hold no keys.

What survives encryption is metadata: packet sizes, inter-arrival timing, direction ratio, burstiness. A video stream and a chat session leave different shapes even when both are opaque.

13 features per flow, RandomForest against XGBoost, selected on validation macro-F1.

Result

macro-F1 0.98 on held-out real captures, reported as a confusion matrix.

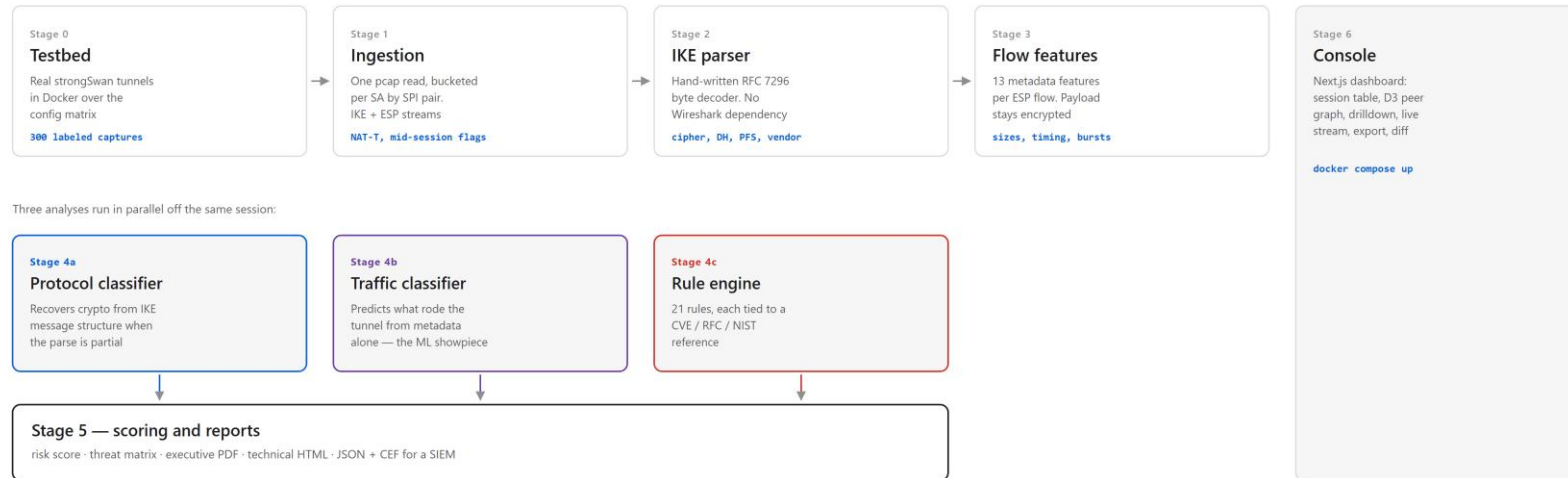
Keeping these apart is the design decision: over-modelling a solved parsing problem is how tools end up with confident, wrong answers.

Covers every item in the brief

- (a) testbed (b) capture (c) protocol and traffic identification
(d) security assessment (e) scores and reports

VaultScope

One capture in, an assessed and scored session record out.



300 labeled captures · 21 security rules · traffic classifier macro-F1 0.98 on held-out real captures

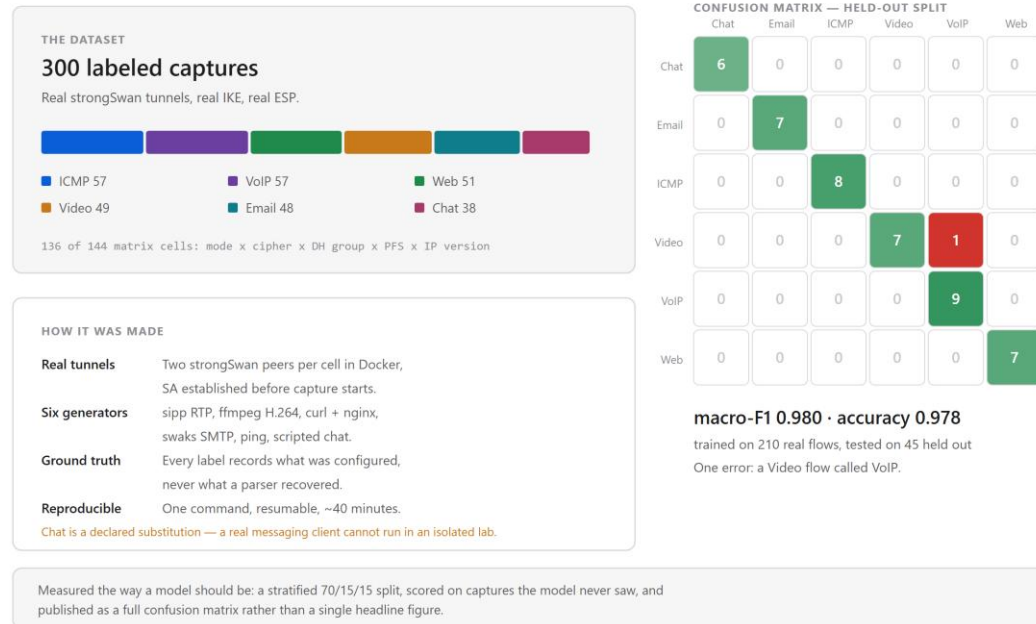
Stack

- Python 3.11, scapy, FastAPI, SQLite
- scikit-learn RandomForest and XGBoost, picked by macro-F1
- strongSwan in Docker; tcpdump / dumpcap capture
- Jinja2 + WeasyPrint reports
- Next.js, Tailwind, Recharts, D3
- One docker compose command

- **IKE parser:** pure-struct decoder. Aggressive Mode comes from the ID payload in message 1, not message count; PFS only from a real second DH exchange.
- **Traffic model:** 13 metadata features per ESP flow, trained on 300 real captures, with per-prediction explanations and an abstain threshold.
- **Assessment:** 21 YAML rules. Fields hidden by encryption read 'unknown' or 'inferred', never a guess; imported swanctl / ip xfrm state can fill them.

The dataset is ours, and so is the evidence

No public corpus fits IPsec ESP with per-tunnel crypto labels, so we generated one.



Built and tested

- Runs today: one docker compose command; 380 Python, 57 unit and 23 end-to-end tests in CI
- Real data: 300 captures, 136 of 144 matrix cells, tunnel and transport, IPv4 and IPv6, PFS on and off
- Fast: about 1 s warm for a 2,884-packet, six-tunnel capture
- Honest accuracy: macro-F1 0.98 on 45 held-out flows is an upper bound, not a field number

Constraints and mitigations

- Encrypted payload: metadata only, with a confidence score and an abstain threshold
- Mode, PFS and auth sit in encrypted IKE_AUTH: reported 'unknown' or 'inferred', or filled from gateway state
- Lab-clean, one class per capture: disclosed; held-out runs by cipher and DH group are scripted

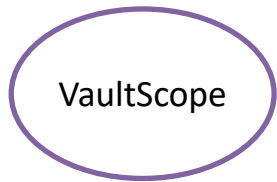
(a) Testbed
done

(b) Capture
done

(c) Identify
done

(d) Assess
done

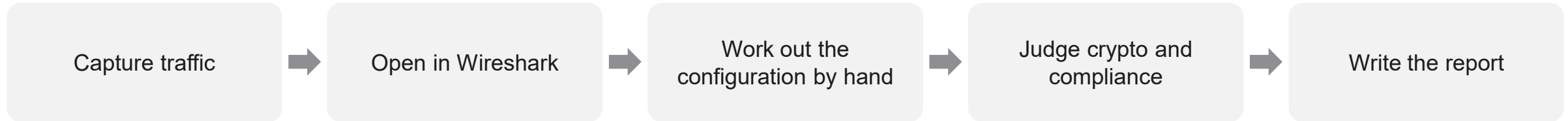
(e) Reports
done



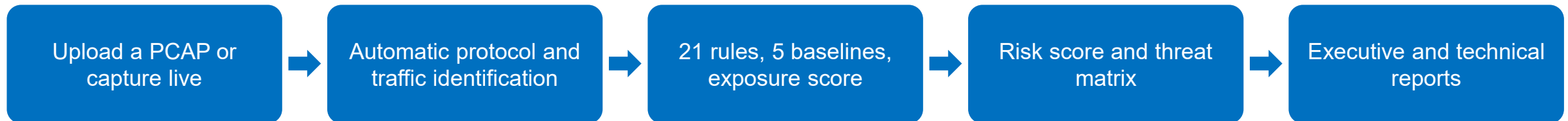
IMPACT AND BENEFITS



TODAY: manual, expert-only



WITH VAULTSCOPE: about 1 s per 2,884-packet capture, warm



- **Analysts:** a scored, referenced verdict on any IPsec capture without reading packets. Every finding names the CVE or RFC behind it and a vendor-specific configuration fix.
- **Auditors and leadership:** compliance mapped to NIST, BSI, CNSA 2.0, CERT-In and TEC ITSAR, a plain-English executive PDF, and SIEM-ready JSON/CEF export.
- **The ecosystem:** the testbed and the 300-capture labeled dataset are reproducible, so others can extend and benchmark the work. Passive by default, so live traffic is never disrupted.

RESEARCH AND REFERENCES



- **Standards:** RFC 7296 (IKEv2), RFC 2408 and 2409 (ISAKMP, IKEv1), RFC 4301 and 4303 (IPsec architecture, ESP anti-replay), RFC 3947 (NAT-T), RFC 7427, RFC 8247 (IKEv2 algorithm guidance), RFC 9370 (multiple key exchanges, post-quantum hybrid).
- **Guidance:** NIST SP 800-77r1, NIST SP 800-131A, NIST IR 8547, BSI TR-02102-3, NSA CNSA 2.0 and CNSSP-15, CERT-In and TEC ITSAR (indicative mappings).
- **Vulnerabilities:** CVE-2016-2183 (SWEET32, 3DES), CVE-2016-1287 (Cisco ASA IKE fragmentation), plus LOGJAM and SLOTH, all covered by the rule engine.
- **Prior art compared:** ike-scan, ikeforce, iker.py, Wireshark, Tufin and AlgoSec, Nessus IPsec plugins. These decode or scan configuration; none pairs runtime attack detection with traffic-type inference inside ESP.
- **Method:** encrypted-traffic classification from packet-size and timing side channels (RandomForest, XGBoost); strongSwan documentation.
- **Source and data:** github.com/shivansh193/vaultscope (README, design document, data/DATASHEET.md).